Lecture 8 recap

In Lecture 8:

- Learnt a five-step process to calculate power series solutions to ODEs
  1. Substitute power series into ODE
  2. Match powers of t
  3. Make infinite sums run over same range, paying attention to small index terms
  4. Form recurrence relation
  5. Check number of linearly independent solutions
- Applied method to example problems

Lecture 9

In Lecture 9, we will:

1. Find a power series solution to Bessel's equation
2. Learn the importance of uniform convergence and the Weierstrass test
3. Define *formal* and *genuine* power series solutions







Case study 3: Transmission lines and waveguides

- We see electricity transmission or power lines everywhere everyday, but what are waveguides? Well they are basically devices that 'guide' waves, like sound, transmitting them in one direction with minimal loss of energy.
- They are typically hollow and have a rectangular or circular cross section. Modelling the latter leads to a **differential equation that can be solved using the power series method. The solutions are expressed in terms of so-called Bessel's functions.** (Gómez-Correa et al (2017) American Journal of Physics)

Bessel differential equation

Order zero Bessel equation

- Another 2nd-order, homogeneous linear ODE

$$r^2 y''(r) + r y'(r) + r^2 y(r) = 0$$

- Arises when studying systems with cylindrical or spherical symmetry
- No solutions in terms of elementary functions (similar to Airy equation)

- Bessel's equation in standard form is

$$y''(r) + \frac{1}{r} y'(r) + y(r) = 0 \, .$$

- $1/r$ is NOT continuous on any interval that includes $r = 0$
- ODE does not satisfy requirements for existence and uniqueness of solutions

Exercise 2.8: Find power series solution to Bessel eqn of order zero

Substitute power series of form $y(r) = \sum_{n=0}^{\infty} c_n r^n$ into ODE (standard form)

$$\sum_{n=0}^{\infty} c_n n(n-1) r^{n-2} + \sum_{n=0}^{\infty} c_n n \frac{1}{r} r^{n-1} + \sum_{n=0}^{\infty} c_n r^n = 0$$

$$\Rightarrow \quad \sum_{m=-2}^{\infty} c_{m+2}(m+2)(m+1) r^m + \sum_{m=-2}^{\infty} c_{m+2}(m+2) r^m + \sum_{m=0}^{\infty} c_m r^m = 0$$

$$\Rightarrow \quad \frac{c_1}{r} + \sum_{m=0}^{\infty} \left\{ c_{m+2}(m+2)^2 + c_m \right\} r^m = 0$$

Therefore, $c_1 = 0$ and $c_{m+2} = -\dfrac{c_m}{(m+2)^2}$ for $m \geqslant 0$.

$$c_2 = -\frac{c_0}{2^2}\,, \quad c_4 = -\frac{c_2}{4^2} = \frac{c_0}{2^2 \cdot 4^2}\,, \ldots, \quad c_{2k} = \frac{(-1)^k c_0}{2^2 \cdot 4^2 \cdots (2k)^2} = \frac{(-1)^k c_0}{(2^k k!)^2}$$

$$c_3 = -\frac{c_1}{3^2} = 0\,, \quad c_5 = 0\,, \ldots, \quad c_{2k+1} = 0$$

The power series method only found one linearly independent solution, which is

$$y_1(r) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2^k k!)^2} r^{2k} = 1 - \frac{r^2}{4} + \ldots \equiv J_0(r)$$

J_0 is the Bessel function of the first kind of order zero.

Bessel functions of higher order and other special functions will be discussed in MAST30031 Methods of Mathematical Physics

Exercise 2.9: Find general solution of Bessel eqn of order zero

Use reduction of order $y(r) = u(r)\, v(r)$ with $v = J_0(r)$

For the Bessel eqn $f_1(r) = 1/r$, and so the reduction of order integrating factor is

$$I = v^2 \mathrm{e}^{\int \frac{1}{r}\mathrm{d}r} = r\, v^2$$

Reduction of order requires we find $w = u'$ from

$$\frac{\mathrm{d}}{\mathrm{d}r}[I\, w] = \frac{I\, g}{v} \qquad \Rightarrow \qquad r\, v^2 u' = C_2 \qquad \Rightarrow \qquad u = C_1 + \int \frac{C_2}{r v^2}\, \mathrm{d}r$$

We can't evaluate the integral, and so we just write

$$y(r) = C_1\, J_0(r) + C_2 \underbrace{J_0(r) \int \frac{1}{r J_0^2}\, \mathrm{d}r}_{y_2(r)}$$

Differentiating under the summation

- The method relies on differentiating the power series term by term
- Is this valid?

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{n=0}^{\infty} c_n t^n \stackrel{?}{=} \sum_{n=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}\, c_n t^n$$

Definition (Uniform convergence)

Sequence $W_N(t)$ converges *uniformly* to $W(t)$ on $t \in D$ if

$$\lim_{N\to\infty} \max_{t\in D} \left|W_N(t) - W(t)\right| = 0$$

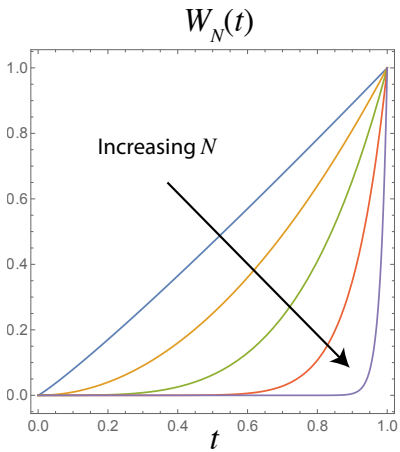
This means the *maximal** error on the domain D goes to zero as $N \to \infty$.

Stronger condition than pointwise convergence!

* Strictly, max should be sup, which you will see if you take MAST20026 Real Analysis.

Exercise 2.10: Consider the sequence $W_N(t) = t^N$ on $D = [0, 1]$

What does the sequence converge to and does it converge uniformly?



W_N converges pointwise to the function

$$W(t) = \begin{cases} 0\,, & 0 \leqslant t < 1 \\ 1\,, & t = 1 \end{cases}$$

$$\lim_{t \to 1^-} W_N(t) = 1 \quad \text{and} \quad \lim_{t \to 1^-} W(t) = 0$$

$$\Rightarrow \quad \lim_{N \to \infty} \max_{t \in D} \big| W_N(t) - W(t) \big| = 1$$

for all N.

Therefore, the sequence $W_N(t)$ does not converge uniformly to $W(t)$ on D.

Definition (Uniform convergence of a series)

- Let $W_N(t) = \sum_{n=1}^{N} w_n(t)$ converge uniformly to $W(t)$ on D, i.e.,
$$\lim_{N\to\infty} \max_{t\in D} \left| \sum_{n=1}^{N} w_n(t) - W(t) \right| = 0$$
- We then say that $W(t) = \sum_{n=1}^{\infty} w_n(t)$ converges uniformly on D

Theorem (Weierstrass test for uniform convergence)

- Let $w_n(t)$ $(n = 1, 2, 3, \ldots)$ be functions on D
- Let $\max_{t\in D} \left| w_n(t) \right| \leqslant a_n$ for some $a_n \in \mathbb{R}$ such that $\sum_{n=1}^{\infty} a_n$ converges
- Then $\sum_{n=1}^{\infty} w_n(t)$ converges *uniformly* on D

Exercise 2.11: Does $\sum_{n=0}^{\infty} \frac{t^n}{n!}$ converge uniformly on $t \in [-b, b]$?

Let $a_n = \frac{b^n}{n!}$ and so $\max_{t \in D} \left| \frac{t^n}{n!} \right| \leqslant a_n$

$\sum_{n=0}^{\infty} a_n$ converges by the ratio test:

$$\lim_{n \to \infty} \left| \frac{b^{n+1}}{(n+1)!} \frac{n!}{b^n} \right| = \lim_{n \to \infty} \left| \frac{\frac{b}{n}}{1 + \frac{1}{n}} \right| = 0 < 1$$

Therefore, $\sum_{n=0}^{\infty} \frac{t^n}{n!}$ converges uniformly on D by the Weierstrass test

- Let $P(t) = \sum_{n=0}^{\infty} w_n(t)$ where w_n are continuous differentiable functions on $D = [a, b]$
- Suppose $P(t)$ converges uniformly on D
- Suppose $\sum_{n=0}^{\infty} w_n'(t)$ converges uniformly on D

Theorem (Differentiating a series)

Then $P(t)$ can be differentiated term by term, i.e.,

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_{n=0}^{\infty} w_n(t) = \sum_{n=0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t} w_n(t)$$

Theorem (Uniform convergence of a power series)

Every power series converges uniformly on every closed interval that lies strictly within its interval of convergence.

Formal and genuine solutions

1. Formal solution: term by term differentiated series satisfies the ODE.
   - Series and its derivatives may or may not converge (either pointwise or uniformly).
   - Series solution may or may not satisfy the ODE.

2. Genuine solution: series and all derivatives up to order of ODE converge uniformly.
   - Term by term differentiation up to the order of the ODE is allowed.
   - Solution satisfies the ODE.

Power series and uniform convergence

Search for a *formal* power series solution to an ODE

$$y(t) = \sum_{n=0}^{\infty} c_n t^n$$

and find coefficients so that the term by term series satisfies the ODE.

If the radius of convergence is non-zero then, for any domain strictly within the radius of convergence, the power series

1. converges uniformly,
2. can be differentiated term by term and
3. is a *genuine solution* to the ODE.